5.3.2 Integration Worksheet

1. Evaluate $\int\left(6t^2+5e^t+2\right)dt$

2. Evaluate $\int\left[10x^4-2\sec^2\left(x\right)\right]dx$

3. Evaluate $\int_{1/2}^{\sqrt{2}/2}-\left(1-x^2\right)^{-1/2}\left[\sqrt{1-x^2}+1\right]dx$

4. Evaluate $\int \left(3x - 10\sqrt{x}\right)^2 dx$.

5. Evaluate $\int_{-1}^{1} \frac{x^4 + 3}{x^2 + 1} dx$. *(Hint: Try to use a fancy 0!)*

6. Evaluate $\int \left(\frac{6x - x^3}{2x^2}\right) dx$.

7. Evaluate $\int_{0}^{2\pi} \tan(x)\,dx$.

8. Evaluate $\int \frac{4\sin^5(x) - 3\left[1-\cos^2(x)\right]}{\sin^4(x)}\,dx$.

9. Evaluate $\int_{\pi/6}^{\pi/3} \left[\csc(x) - \csc(x)\cos^2(x)\right]dx$.

10. Evaluate $\int \frac{\cos(2\theta)+\sin^2(\theta)}{\cos(\theta)}\,d\theta$.

11. Determine the average value of the function $f(x)=\frac{8}{x^2}$ on the interval $[1,5]$.

12. Evaluate $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$.